

# SAIF UR RAHMAN KHAN

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## EDUCATION

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| <b>Jamia Millia Islamia (JMI), New Delhi, India</b><br>Bachelor of Technology (Electrical Engineering)                       | Nov 2021 - May 2025<br>CGPA- 8.5/10<br><b>Top 10%</b> (First Division with honors) |
| <b>RD Public School, Betul, Madhya Pradesh</b><br>Class 12 <sup>th</sup> (Senior Secondary), Physics  Chemistry  Mathematics | Apr 2020 - May 2021<br>Percentage - 94.8%                                          |
| <b>RD Public School, Betul, Madhya Pradesh</b><br>Class 10 <sup>th</sup> (Matriculation)                                     | Apr 2018 - Feb 2019<br>Percentage - 92.2%                                          |

## RESEARCH INTERESTS

Renewable Energy, E-mobility, Digital Twins for Hybrid Micro-grids, Off-grid Energy Systems, Techno-economic Study of Hybrid Renewable Energy Systems, Vehicle-to-X

## RELEVANT COURSEWORK

Engineering Economics and Industrial Management, Renewable Energy Sources, Utilization of electrical energy, Power Electronics, Power System analysis, Electrical power generation, Advanced Control system, Supervisory Control and Data Acquisition (SCADA) & Smart Grid Technologies, Power system operation and control, Programing languages, Soft Computing

## RESEARCH EXPERIENCE

**Jamia Millia Islamia University, New Delhi** May 2023 - May 2025  
**Smart Energy Lab, Supervisor:** Prof. Majid Jamil  
**Undergraduate Thesis** – Design and Development of Hydrogen Fuel Cell Based Electric Bicycle.

### Key roles:

- Designed and developed a hydrogen PEMFC-hybrid electric bicycle by integrating a 150W Ecosense PEM fuel cell with a 36V/10.4Ah Li-ion battery and 350W BLDC hub motor, extending the hybrid range up to 100 – 120 km from a baseline of 25–30 km
- Analyzed PEM fuel cell V-I characteristics to identify the Maximum Power Point (MPP), optimizing energy extraction efficiency for the integrated hybrid powertrain
- Integrated a DC-DC converter circuit (36V → 12V) to manage the hybrid charging profile between the fuel cell and lithium-ion battery
- Fabricated the e-bike prototype by modifying a standard pedal bicycle frame, integrating the wiring harness, throttle, cruise control, real-time SOC display, lights, and horn
- Modelled a 3D SolidWorks replica of the hydrogen fuel cell bicycle to validate compact component placement and structural layout
- Secured ₹20,000 in project funding from the Jamia Engineering Alumni Association for prototype hardware procurement

**Jamia Millia Islamia University, New Delhi** May 2023 - May 2025  
**Supervisor:** Prof. Majid Jamil

**Undergraduate Research Project** - Techno-Economic Analysis of Off-Grid Hybrid Renewable Energy-Based Hydrogen Refueling and Power Supply for Remote Indian Villages

### Key roles:

- Modelled site-specific load demand profiles and renewable resource data (solar irradiance, wind speed) for 3 remote, unelectrified Indian villages in Madhya Pradesh, using real grid-connectivity data.

- Simulated and compared 4 hybrid RE configurations in HOMER Pro (PV+WT+DG+Bat, PV+WT+DG, PV+WT+Bat, PV+WT+HFC), selecting PV+Wind+Hydrogen Fuel Cell as the optimal net-zero, zero-LPSP configuration for all sites
- Achieved Levelized Cost of Energy of \$1.25/kWh and LCOH of \$3.22/kg with an 80kW PEM electrolyzer + 250kW fuel cell replacing conventional BESS, enabling both power supply and hydrogen refueling from a single system.
- Extended findings into a journal paper currently under writing: "Techno-Economic Analysis of Off-Grid Hybrid Renewable Energy-Based Hydrogen Refueling and Power Supply for Remote Indian Villages" (Co-author: Prof. Majid Jamil)

**Jamia Millia Islamia University, New Delhi**

May 2023 - May 2025

**Supervisor:** Prof. Majid Jamil

**Undergraduate Research Project** - Industrial-Scale Green Hydrogen Production Facility (SIH 2024 Hardware Finals)\_Team Hydro | Ministry of Power, Govt. Of India, PS Code: 1574

**Key roles:**

- Designed a >50 TPD green hydrogen facility prototype at Ramagundam, Telangana, integrating 530 MW hybrid RE (130MW solar + 400MW wind) with 80 SOEC electrolyzer units targeting LCOH < \$2/kg
- Proposed sustainable ammonia via Haber-Bosch as primary storage (300× cheaper lifetime than compressed H<sub>2</sub>) and integrated Solhyd panels + low-head water turbine for intermittency compensation
- Built a lab-scale prototype with CSP-driven electrolysis and a 3-stage water purification train (distillation → UV → carbon chamber), presented at SIH 2024 Hardware Finals, KIET Ghaziabad

**Jamia Millia Islamia University, New Delhi**

May 2023 - May 2025

**Supervisor:** Prof. Majid Jamil

**Undergraduate Minor Project** - IoT-Based Solar Powered Pumping and Manuring System (*DTU-IIF IdeaThon 2024 Winner*) Team Agrivolters | JMI

**Key roles:**

- Designed an off-grid solar irrigation system (NPK + moisture sensors, Arduino, GSM, 3HP pump; CAPEX ₹2.2L) automating watering and manure delivery based on real-time soil data
- Modelled a shared-use revenue model (₹8k–10k/month from neighboring farms) enabling farmer ROI in 18–25 months; won ₹10,000 prize at DTU-IIF IdeaThon 2024

## WORK EXPERIENCE

**Denso International India Ltd, Manesar, Gurgaon, Haryana**

Jul 2025 - Present

**Position:** R&D Trainee

**Key roles:**

- Benchmarked Valeo India's 3-in-1 combo box (PDU + DC-DC Converter + OBC assembly) against Denso's internal design targets, evaluating parameters including power density, thermal performance, and integration architecture.
- Analyzed the impact of Vehicle-to-X (V2X) bi-directional power flow on the switching stress, thermal loading, and reliability of On-Board Charger (OBC) and DC-DC converter power modules in the X-in-1 by Denso Japan.
- Designed current sensor mounting configurations and battery pack structural parts, ensuring compatibility with HEV/BEV platform requirements and safety margins.
- Reviewed and ensured compliance of HEV/BEV electrical systems with Indian automotive standards including AIS-038, IS 17017, and IS 17107.
- Validated current sensor performance through on-site testing at customer facilities including Maruti Suzuki India Ltd (MSIL) and Toyota, documenting measurement results and deviations.

## PUBLICATIONS

Khan, S. U. R., Ahmad, A., Jamil, M., & Irfan, A. (2025). State-of-charge estimation of Li-Ion battery in electric vehicles using deep neural networks. In *2025 International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI)* (pp. 861–865) IEEE .(Conference Paper)

<https://doi.org/10.1109/IC3ECSBHI63591.2025.10990906>

## INTERNSHIPS

**Jamia Millia Islamia, New Delhi**

Mar 2024-Jun 2024

**Position:** Research Intern**Control Lab, Supervisor:** Dr. Arunesh Kumar Singh**Project** – AI based Smart Eddy Current Energy Absorbing System**Key roles:**

- Developed hardware prototype models of an Eddy Current Braking System (ECBS) using both permanent magnets and electromagnets to study contactless energy absorption mechanisms
- Implemented Fuzzy Logic Controllers (FLC) in the simulation model to handle system uncertainties and improve robustness of the braking control response
- Applied AI-based smart analysis to perform real-time decision-making and optimize the rate of energy absorption based on speed and load conditions
- Investigated the application of eddy current-based Energy Absorbing Systems (EAS) as a replacement for conventional hydraulic absorbers in Aircraft Arrestor Barrier Systems (AABS). Validated experimentally that braking torque is directly proportional to induced eddy current, confirming theoretical models referenced earlier.

**Tata Power DDL, New Delhi**

Jun 2024 - Jul 2024

**Position:** Summer Research Intern**Project:** Predictive Model for Smart Meters integrated with Rooftop Photovoltaic**Key roles:**

- Collected and pre-processed historical smart meter data including hourly load profiles, voltage levels, and energy consumption patterns across residential distribution feeders
- Developed a machine learning-based predictive model to forecast net grid consumption by integrating rooftop PV generation data, improving demand forecasting accuracy
- Analyzed the impact of rooftop solar penetration on feeder-level voltage profiles and reverse power flow conditions in the Delhi distribution network
- Visualized key performance indicators including generation vs. consumption trends, peak shaving potential, and PV self-consumption ratio using Python-based data dashboards

**Tesla Transformers India Ltd, Bhopal**

May 2024 - Jul 2024

**Position:** Summer Intern**Key roles:**

- Studied the design and performance characteristics of Inverter Duty Transformers (IDTs), analyzing K-factor ratings, harmonic distortion tolerance, and thermal derating under non-sinusoidal load conditions
- Explored Agrivoltaics system configurations, examining the co-location of solar PV panels with agricultural land to evaluate energy yield and shading impact on crop productivity
- Processed and visualized field measurement data using Python and Excel, generating comparative performance plots for transformer efficiency, harmonic spectrum, and PV output variation
- Assisted in site measurements and documentation of transformer nameplate parameters, winding configurations, and insulation test records

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**SKILLS****Programming:** Python (NumPy, Pandas, Matplotlib, TensorFlow/Keras), MATLAB, Arduino/ESP32 (embedded C)**Hardware:** BLDC Hub Motors, PEM Fuel Cells, Li-ion Battery Systems, DC-DC Buck Converters, On-Board Charger (OBC) & PDU Systems, Current Sensors, IoT Sensors & Microcontrollers, Electrolyzer Systems (SOEC/PEMFC), Eddy Current Braking Systems, Inverter Duty Transformers**Software:** Siemens NX, AutoCAD Electrical, Denso internal Automotive Design & BOM Documentation Tools, HOMER Pro (Hybrid Energy Optimization), SolidWorks (3D Modelling), MATLAB Simulink, LaTeX, PVSyst, MS Office Suite, Google Collab, GitHub, Power BI, SQL

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**CERTIFICATIONS**

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| <b>Hydrogen for the Energy Transition: Power-to-X</b><br>WWF Germany (WWF Akademie) — Online<br>Skills: Power-to-X (PtX), Fischer-Tropsch Synthesis, Green Hydrogen             | Mar 2025 |
| <b>Renewable Energy Technologies</b><br>ASSET EnergyTransition — Online<br>Skills: Renewable Energy Systems, Solar Energy, Energy Transition                                    | Oct 2024 |
| <b>Learning SCADA: Collect, Analyze, and Visualize Data for Industrial Automation</b><br>LinkedIn Learning — Online<br>Skills: SCADA, Industrial Automation, Data Visualization | Sep 2024 |

LEADERSHIP EXPERIENCE / EXTRA CURRICULAR ACTIVITIES

- Chairperson, Power and Energy IEEE Sub-Society** - JMI Student Branch, New Delhi - Led the Power & Energy sub-society, organizing technical sessions, workshops, and industry talks for undergraduate students. Feb 2024 – Feb 2025
- R&D Team Member** - ENACTUS Jamia Millia Islamia - Contributed to health-focused social entrepreneurship projects, combining engineering solutions with community outreach under the ENACTUS framework. Jun 2024 – May 2025
- Field Volunteer** — Aashman Foundation - Actively volunteered in social service initiatives, supporting underprivileged communities through field-level outreach programmers over 2+ years. Aug 2023 – Mar 2026
- Solar Charging Seminar** - Delivered a seminar on Solar Charging Stations for EVs, reviewing grid-tied and off-grid architectures, battery swapping methods, and proposing a model framework covering PV sizing, storage, grid integration, and safety compliance for Indian EV infrastructure. May 2024

AWARDS /ACHIEVEMENTS

- Winner, DTU-IIF IdeaThon 2024** - Won ₹10,000 prize for an Agri-Voltaic solar-powered IoT irrigation and manuring system project. Aug 2024
- SIH 2024 Hardware Edition Finalist** - Selected for Smart India Hackathon 5-days finale (Hardware Edition) at KIET Ghaziabad, U.P. for the Industrial-Scale Green Hydrogen Production Facility project. Dec 2024
- Project Funding Awardee** - Received ₹20,000 from Jamia Engineering Alumni Association for the Hydrogen Fuel Cell E-Bike major project Mar 2025

REFERENCES

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| Prof. Majid Jamil<br>Professor<br>Electrical Engineering<br>Jamia Millia Islamia<br>New Delhi , India<br><a href="mailto:mjamil@jmi.ac.in">mjamil@jmi.ac.in</a><br>+91 9313462108 | Prof. Shakeb Ahmad Khan<br>Professor<br>Electrical Engineering<br>Jamia Millia Islamia<br>New Delhi , India<br><a href="mailto:skhan3@jmi.ac.in">skhan3@jmi.ac.in</a><br>+91 9650087393 | Prof. Arunesh Kumar Singh<br>Associate Professor<br>Electrical Engineering<br>Jamia Millia Islamia<br>New Delhi ., India<br><a href="mailto:asingh1@jmi.ac.in">asingh1@jmi.ac.in</a><br>+91 9911889119 |
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